

Score-Based Generative Diffusion Models for Dynamic Objects

Thesis Direction Notes

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Abstract

Concerning the statistical-physics analysis of score-based diffusion models applied to dynamic data (e.g. videos or synthetic trajectories). We aim to model and understand the evolution of score functions across time in systems with simple, known physical dynamics.

1 Problem Statement: Score-Based Diffusion Models for Dynamic Objects

We study generative diffusion models not for static data (images) but for *dynamic objects*, i.e. short sequences of frames forming a trajectory. Consider a rotating digit μ from the MNIST dataset. The rotation is a sequence index by u :

$$x^\mu = \{x_u^\mu\}_{u=0}^{T-1}, \quad x_u^\mu \in \mathbb{R}^d,$$

which equates to a trajectory running for T frames, each in $\mathbb{R}^d$.

Stupid approach. Each trajectory can be seen as living in $\mathbb{R}^{dT}$. This is inefficient because the space grows but information stays the same, as the object is just rotated.

Smart approach. Given high correlation between consecutive frames, investigate the mapping between the score functions at two rotation indices. Crucially, we aim to exploit some structure across *internal time* u . Mathematically, we have:

$$s_{i,u}(\cdot) \simeq f(s_{i,u-1}(\cdot))$$

where

$$s_\phi(x, u, t) \approx \nabla_x \log p_t(x \mid u),$$

the score of each frame at diffusion time t .

2 A Simple Physical Model: Motion on a Circle

To have a fully solvable setting, we consider a 2D stochastic process on a circle:

$$(x(u), y(u)) = (r(u) \cos \theta(u), r(u) \sin \theta(u)).$$

The internal time $u \in [0, 2\pi/\omega]$ is sampled at T discrete frames. We impose radial relaxation + angular drift via Langevin SDEs:

$$dr(u) = (1 - r(u)) du + \sqrt{2T_r} dB_r(u), \tag{1}$$

$$d\theta(u) = \omega du + \sqrt{2T_\theta} dB_\theta(u). \tag{2}$$

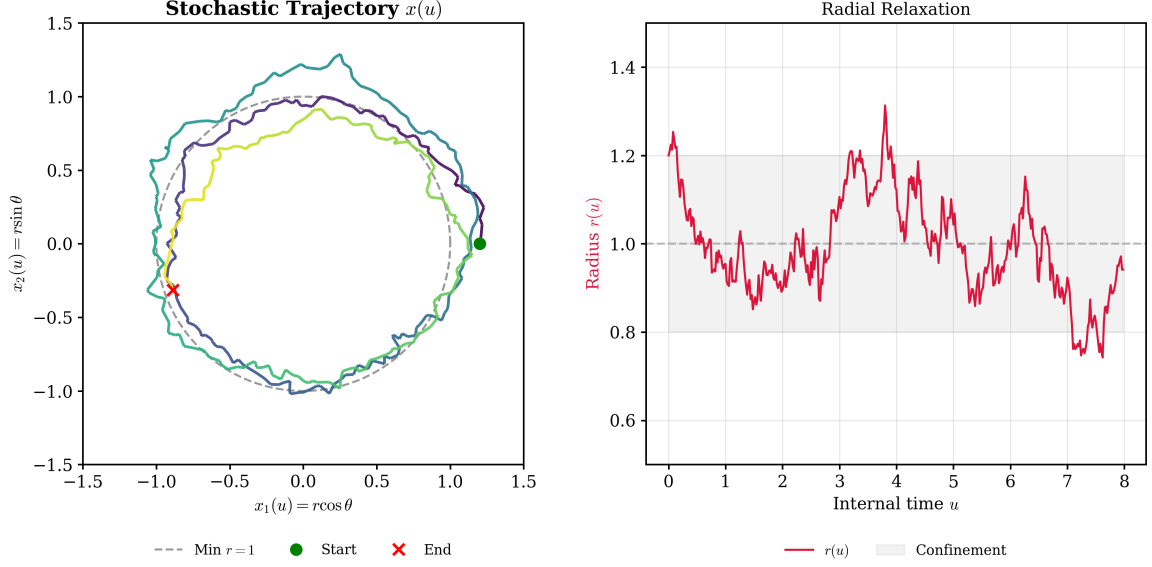


Figure 1: Stochastic dynamics on the ring. Left: trajectory $x(u)$ in the (x_1, x_2) plane, fluctuating around the circle $r = 1$, with start and end points highlighted. Right: corresponding radial evolution $r(u)$, showing relaxation towards the confining radius $r = 1$ under the SDE $dr = (1 - r) du + \sqrt{2T_r} dB_r$.

Thus, each point of the trajectory lies inside a convex radian confinement potential with minimum at radius 1, visualized in Figure 3, with equation:

$$V(r) = \frac{1}{2}(r - 1)^2$$

Angular motion is almost deterministic, with small noise. A single trajectory drawn from this process yields our data:

$$\mathcal{D} = a^\mu = \{x^\mu(u), y^\mu(u)\}, \quad u = 1, \dots, T - 1, \mu = 1, 2, \dots, |\mathcal{D}|$$

This toy model already has:

- continuous motion,
- curvature,
- a known generator of dynamics,
- nontrivial but analyzable structure.

3 Forward Diffusion Process per Frame

To build a score-based diffusion model, each frame is corrupted independently using the Ornstein–Uhlenbeck process:

$$X_t^\mu = a^\mu e^{-t} + \sqrt{1 - e^{-2t}} Z^\mu, \quad Z^\mu \sim \mathcal{N}(0, I_d),$$

where $a^\mu = x_u^\mu$ is the clean frame.

Thus the model must learn

$$s_\phi(x, u, t) = \nabla_x \log p_t(x | u).$$

The joint distribution of $\{x_u\}$ is complicated, but the physics of the process implies strong temporal correlations between the true scores across u .

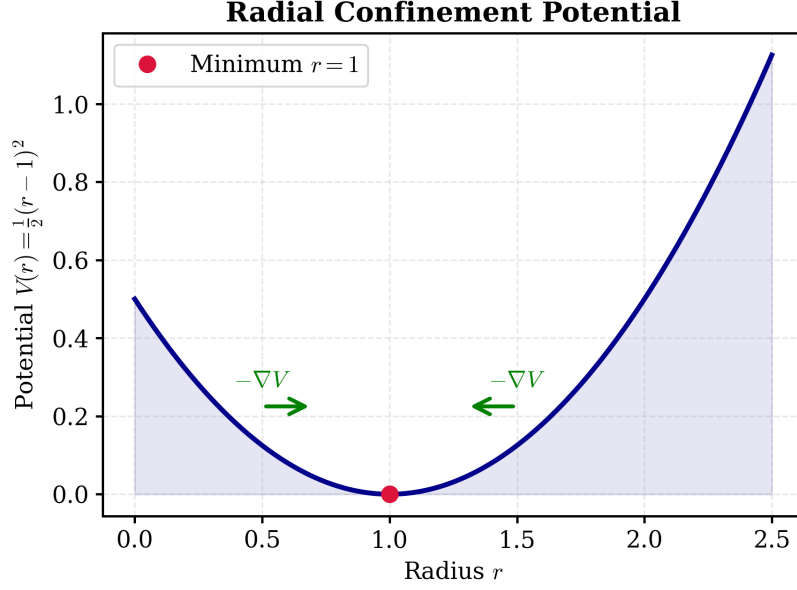


Figure 2: Radial confinement potential $V(r) = \frac{1}{2}(r-1)^2$. The minimum at $r = 1$ defines the preferred radius of the motion, while the arrows indicate the restoring force $-\nabla V$ that pulls trajectories back towards the ring.

4 Temporal Evolution of the Score

Let the score of the full trajectory be:

$$S(x) = \nabla_x \log p(x) \in \mathbb{R}^{dT}, \quad S_{:,u}(x) \in \mathbb{R}^d.$$

Instead of learning all dT degrees of freedom independently, we hypothesize a **propagator** P such that:

$$S_{:,u}(x) \approx P(S_{:,u-1}(x)),$$

where P may be:

- linear in the simplest model,
- stationary (independent of u),
- related to the generator of the underlying SDE.

For rotation, P is nearly a rotation operator acting on the score field:

$$S_{:,u} \approx R_{\Delta} S_{:,u-1}.$$

Learning P means learning the *internal dynamics of the score*. This gives new research directions:

- How much temporal structure can we recover?
- Does the learned P match the true SDE generator?
- How does knowing P improve generation?
- Can P enable extrapolation (predicting future frames)?

5 Diffusion for Video - Shallow Literature Review

There exist several complementary approaches to address video processing and generation using diffusion models. All these methods offer insights into how our research problem can be approached.

5.1 Neural Differential Equations (pre score-based diffusion)

- [Vid-ODE](#). Learns continuous-time dynamics using ODE-ConvGRU as an encoder.
- [Latent Neural Differential Equations for Video Generation](#). Study the effects of Neural Differential Equations to model the temporal dynamics of video generation.

5.2 Foundations

- [Video Diffusion Models \(VDM\)](#). Inter-frame dependencies via 3D convolutions or attention. Propose “gradient guidance” conditioning method during diffusion sampling step.
- [Image Diffusion Models Exhibit Emergent Temporal Propagation in Videos](#). Find that score network of image generation diffusion models implicitly encodes temporal semantics. Implies that temporal coherence is latent in score-based models even when not explicitly trained for video.

5.3 Sequential

- [Masked Conditional Video Diffusion \(MCVD\)](#). Train a conditional score model that takes a set of context frames as input and predicts the next (or missing) frames. State-of-the-art results on standard video prediction and interpolation benchmark. Interestingly, feature a [Stochastic Moving MNIST experiment](#).
- [FramePack](#). Compresses the past frames into a fixed-length representation based on their importance and introduce drift-prevention strategies. [Progressive Autoregressive VDM \(PA-VDM\)](#). Progressive noise scheduler to maintain quality.

5.4 Motion Guidance

- [Motion-Guided Latent Diffusion \(MGLD\)](#) Conditions the score function on learned motion representations by integrating optical flow. At inference, compute optical flow between low-resolution input frames (true motion between frame $u - 1$ and u) and use it to warp the latent features of frame $u - 1$ toward frame u .

5.5 Multi-view Generation

- [Diffusion](#)². Assume that given all other frames and views, each element is conditionally independent. Then, the joint score approximately equals a convex combination of the individual scores for each frame and each view. Illustrates a form of score propagation.

6 Research Questions

We formalize the core question:

Given dynamic trajectories with known or analyzable underlying physics, can diffusion models learn per-frame scores that obey a simple temporal evolution law? Can we identify (or learn) a stationary operator P such that

$$S_{:,u} \approx P(S_{:,u-1})$$

and use this structure for better generation, interpretation, and extrapolation?

This raises several concrete sub-questions:

- Q1.** What is the exact form of the true score for our circle dynamics? Can it be computed analytically or approximated?
- Q2.** Does the sequence of scores $\{S_{:,u}\}$ satisfy an approximate linear recursion?
- Q3.** Is the propagator P time-stationary?
- Q4.** How does diffusion noise affect temporal coherence of the score?
- Q5.** Can the learned P enable prediction beyond the observed frames (“next-token” prediction)?
- Q6.** How does PCA of the data relate to PCA of the score fields?
- Q7.** How does this compare to existing video-diffusion approaches that treat sequences as blocks rather than time-correlated structures?

7 Objectives of the Thesis (to be refined later)

- Derive analytically the ideal score for simple dynamic systems.
- Simulate and visualize true score fields, their rotation, and temporal coherence.
- Train diffusion models on dynamic datasets and extract empirical score fields.
- Fit or learn the propagator P ; test linear vs. nonlinear forms.
- Compare learned P with ground-truth SDE dynamics.
- Analyze extrapolation performance.
- Explore whether this reveals a universal structure for dynamic generative diffusion.